



# Cloud Computing Essentials

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## AWS Compute Services

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## What is compute?

- In cloud computing, the term “compute” describes concepts and objects related to software computation.
- It is a generic term used to reference processing power, memory, networking, storage, and other resources required for the computational success of any program.

## What are compute resources?

- Compute resources are measurable quantities of compute power that can be requested, allocated, and consumed for computing activities.
- Some examples of compute resources include:
  - **CPU** is the brain of any computer. CPU is measured in units called millicores.
  - **Memory** is measured in bytes. Applications can make memory requests that are needed to run efficiently.

## What is cloud computing?

Cloud Computing is a generic term for anything that involves delivering computer resources over the internet.

Users can access large amounts of computing power on demand.

They can purchase by the minute or the hour and use as much or as little as required for their computing activities.

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### What are compute services?

- Compute as a service (CaaS) is a **pay-per-use business model that lets organizations select cloud compute resources on an on-demand basis**.
- CaaS enables tools and computing resources to process data and run apps.
- Compute platforms, such as AWS Compute, supply a virtual server instance and storage and APIs that let users migrate workloads to a virtual machine.
- Users have allocated compute power and can start, stop, access, and configure their computer resources as desired.
- Compute services are also known as Infrastructure-as-a-Service (IaaS).

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## Amazon EC2 instance

- In AWS Compute services, virtual machines are called instances
- An Amazon EC2 (Elastic Compute Cloud) instance is a virtual machine (VM) that provides resizable computing capacity in the cloud.
- AWS EC2 provides various instance types with different configurations of CPU, memory, storage, and networking resources so a user can tailor their compute resources to the needs of their application.

## Deploying Amazon Elastic Compute Cloud Servers

- To get your virtual machine (VM) instance running, you'll first define the elements one at a time.
  - Amazon Machine Image (AMI) – OS and a software stack
  - Instance type – CPU, memory modules, network adaptor
  - Storage volumes – available through the Elastic Block Store (EBS).



## Amazon Machine Images

- An *image* is a software bundle that was built from a template definition and made available within a single AWS Region.
- The bundle can be copied to a freshly created storage volume that will become a bootable drive and turn the VM it's attached to, into a fully operational server.
  - AMIs with clean, cloud-optimized operating systems can be selected
  - OS preloaded with one of hundreds of popular software stacks can be found

## Amazon Machine Images

AMIs are organized into four collections:

- Quick Start set (pre-configured, popular OS) - Best for beginners, offering pre-configured images like Amazon Linux 2023, Ubuntu
- My AMIs (custom images) - Contains your own custom-created images.
- AWS Marketplace (vendor-provided software) - Contains specialized software, including commercial databases or pre-configured web servers, which may have additional fees.
- Community AMIs (shared by others) - Thousands of free or open-source images provided by other users. Note that these are not verified by AWS.

## EC2 Instance Types

There are six types of instances:

- **General purpose instances** provide a balance of compute, memory and networking resources, and can be used for a variety of diverse workloads. These instances are ideal for applications that use these resources in equal proportions such as web servers and code repositories. Example: M7, Mac, M6, M5, M4, T4, T3, T2
- **Compute optimized instances** are used to run high-performance compute applications that require fast network performance, extensive availability, and high input/output (I/O) operations per second. Scientific and financial modeling and simulation, big data, enterprise data warehousing, and business intelligence are examples of this type of application. Example: C7, C6, C5, C4
- **Accelerated computing instances** use hardware accelerators, or co-processors, to perform functions, such as floating point number calculations, graphics processing, or data pattern matching, more efficiently than is possible in software running on CPUs. Example: P4, P3, P2, DL, Trn, Inf, G5, G4, G3, F1, VT

## EC2 Instance Types

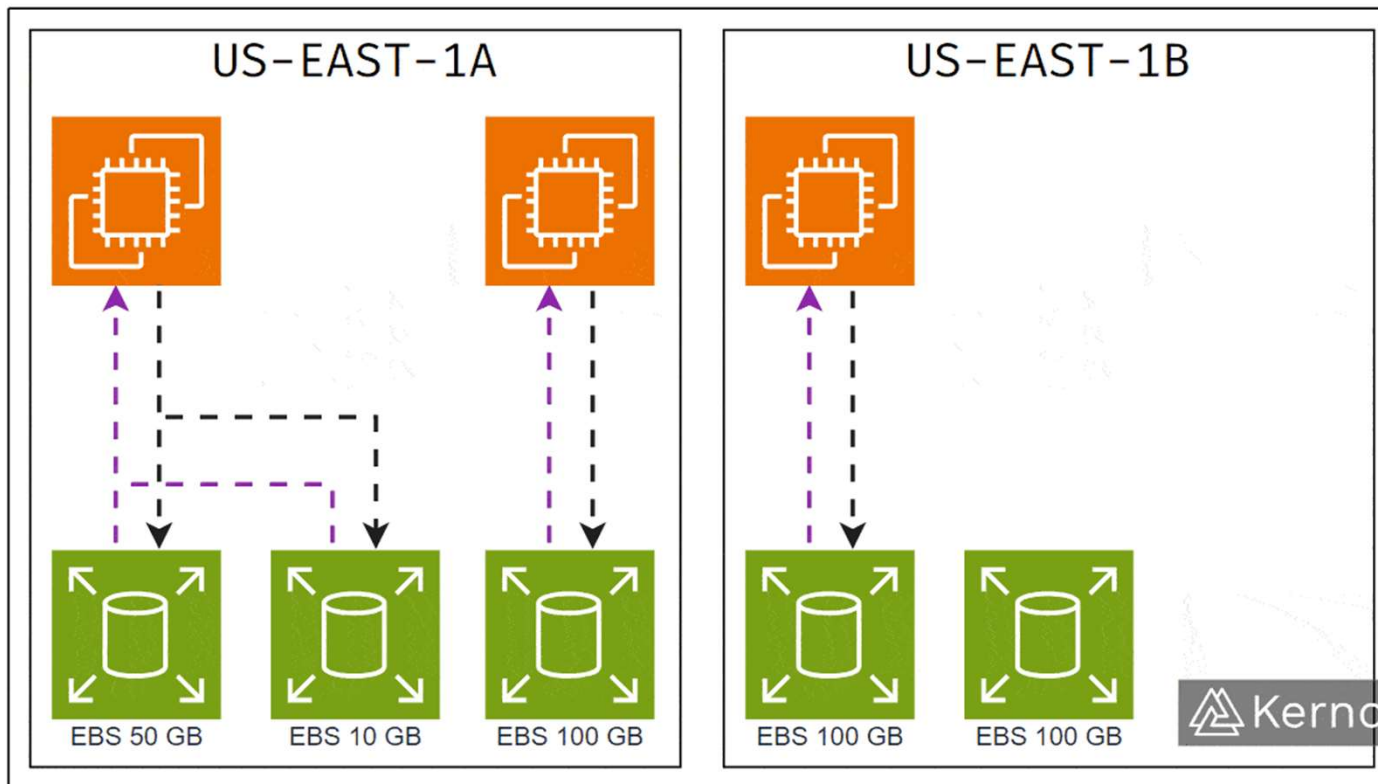
(contd..)

- **Memory optimized instances** use high-speed, solid-state drive infrastructure to provide ultra-fast access to data and deliver high performance. These instances are ideal for applications that require more memory and less CPU power, such as open-source databases and real-time big data analytics. Example: R7, R6, R5, R4 X2, X1, High Memory, z1d
- **Storage optimized instances** are designed for workloads that require high, sequential read and write access to very large data sets on local storage. They are optimized to deliver tens of thousands of low-latency, random I/O operations per second (IOPS) to applications. Example: I4, Im, Is, I3, D2, D3, H1
- **HPC optimized instances** offer the best price performance for running HPC (High-performance computing) workloads at scale. HPC instances are ideal for applications that benefit from high-performance processors such as complex simulations, deep learning, and visual effects rendering. Example: Hpc7, Hpc6

## Server Storage: Amazon Elastic Block Store

- The physical drive where an EBS volume exists may live quite a distance from the physical server that's giving your instance life.
- EBS volumes speak with your instance over a super low- latency network connection spanning the data center.
- EBS makes a trade-off for its slightly slower performance compared to instance store volumes with benefits such as:
  - Persistence - the data will survive shutdowns and system crashes. That can be a factor for workloads where data persistence is necessary.
  - Encryption - EBS volumes can be encrypted, important when working with sensitive data.
  - Flexibility - EBS volumes can be moved around, mounted on other instances, and converted to AMIs.

## Server Storage: Amazon Elastic Block Store

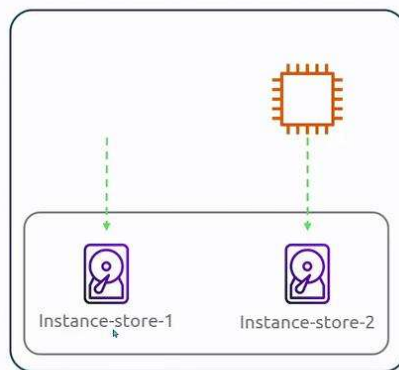


## Server Storage: Amazon EC2 Instance Store Volumes

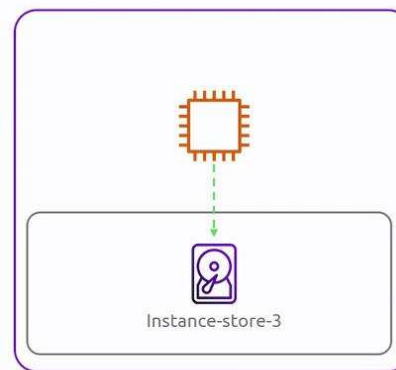
- Instance store volumes are directly attached to the host server, giving them a speed edge over EBS
- It is perfect for ephemeral workloads like real-time data processing
- Instance types can enjoy the benefits of having their data on physical drives attached directly to the physical instance server.
  - The downsides of instance store volumes (ephemeral data, no encryption, and lack of flexibility) are offset by faster data reads and writes.
  - This can be useful for processing and analyzing fast-moving data streams where the data itself doesn't need to be persistent.

## Server Storage: Amazon EC2 Instance Store Volumes

### Instance Storage Architecture



Host 1



Host 2

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## Understanding EC2 Pricing Models

- On- Demand Instances - They are the priciest. Perfect for short-term, unpredictable workloads like that one-day sale.
- Reserved Instances - For uninterrupted usage for over a month, typically sold in 1-3 year terms. a discount card for EC2 instances, not the instance itself. The reservation applies automatically to matching instances in the specified region. Reserved instances are paid for using an All Upfront, Partial Upfront, or No Upfront payment option.
- Spot Instances - For workloads that don't need to run constantly and can survive unexpected shutdowns, AWS makes unused compute capacity available at steep discounts— as much as 90% off the on-demand cost. The capacity can be reclaimed on two minutes' notice.

## What are the benefits of AWS compute services?

AWS Compute services offer the broadest and deepest functionality for compute. Key benefits of using AWS Compute include:

- **Right compute for your workloads**

Amazon EC2 (Amazon Elastic Compute Cloud) offers granular control of application infrastructure. Amazon Elastic Container Services (Amazon ECS) offer choice and flexibility to run containers.

- **Built-in security**

AWS offers significantly more security, compliance, and governance services. The AWS Nitro System has security built in at the chip level.

- **Cost optimization**

With AWS compute you pay only for the instance or resource you need, for as long as you use it, without requiring long-term contracts or complex licensing.

- **Flexibility**

AWS provides multiple ways to build, deploy, and get applications to market quickly. For example, Amazon Lightsail is an easy-to-use service that offers you everything you need to build an application or website.



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## What is a Security Group?

- An Amazon EC2 security group acts as a virtual firewall for your EC2 instances, controlling both inbound (incoming) and outbound (outgoing) traffic at the instance level.
- It regulates network traffic based on rules defined by protocol, port, and source/destination IP address, ensuring only authorized traffic reaches your resources.
- When you create a VPC, it includes a default security group. If you don't specify a security group when launching an instance, the default one is used.

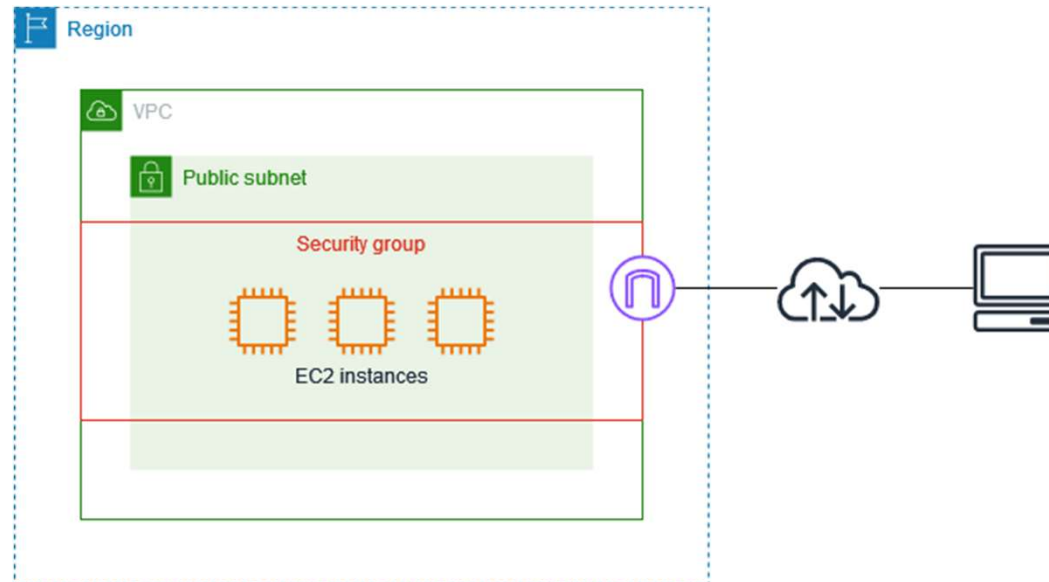
## Key Aspects of EC2 Security Groups:

- **Virtual Firewall:** It controls traffic for one or more instances, acting as a protective barrier.
- **Stateful Filtering:** They are stateful, meaning if an inbound request is permitted, the corresponding outbound response is automatically allowed, regardless of outbound rules.
- **Default Behavior:** By default, all outbound traffic is allowed, while all inbound traffic is blocked until explicitly allowed via rules.
- **Rule Flexibility:** You can add or modify rules (e.g., allow SSH on port 22, HTTP on port 80) at any time, and these changes are automatically applied to associated instances.
- **Instance Level:** They operate at the instance level, not the subnet level, offering granular control.
- **Multiple Groups:** You can associate one or more (upto 5) security groups with a single instance.

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The following diagram shows a VPC with a subnet, an internet gateway, and a security group.



- The subnet contains EC2 instances.
- The security group is associated with the instances.
- The only traffic that reaches the instance is the traffic allowed by the security group rules.





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## AWS Networking Services

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## What is Networking?

- Networking is about transporting data to and from your AWS resources.
- Achieving this depends on many factors, including the type of data, the speed of data transport, your security requirements, who or what will be accessing that data, and how they'll be accessing it.
- The three core networking services AWS offers:
  - Virtual Private Cloud
  - Route 53
  - CloudFront

## Virtual Private Cloud

- A VPC is a logically isolated virtual network in AWS.
  - Default VPCs allow internal instances to access the Internet.
  - Non-default VPCs are isolated, requiring explicit config for external access
- Amazon creates a default VPC in each region when you sign up for an AWS account.
- Key components include
  - subnets (public and private)
  - gateways (internet and NAT) and route tables
  - security provided by security groups and network ACLs.

## **Route 53**

- Route 53 is Amazon's global Domain Name System (DNS) service.
- The primary purpose of DNS is to translate human-readable domain names (such as example.com) into IP addresses.
  - User's system sends out a query to its configured DNS server asking for the IP address of that domain.
  - The DNS server then sends the query to the domain's authoritative DNS server—the one that's in charge of the example.com domain name.
  - The authoritative DNS server responds with the IP address for example.com.
- This process of translating a domain name to an IP address is called name resolution.

## Cloud Front

- Amazon CloudFront is a CDN that speeds up content delivery by caching it in 150+ edge locations worldwide on six continents, reducing latency and increasing availability
- To deliver content via CloudFront, you create a distribution (Web or RTMP)
  - Web distribution: for static/dynamic content (HTTP/HTTPS), origin can be a web server or public S3 bucket.
  - RTMP distribution: for streaming media, requires media player and files in S3

## Virtual Private Cloud – Concepts and Components

- Subnets
  - VPC CIDR Blocks
- Internet Access
  - Internet Gateway
  - NAT gateway
- Security
  - Security Groups
  - Network Access Control Lists



## Virtual Private Cloud – CIDR Blocks

- CIDR (Classless Inter-Domain Routing) is a method to denote IP address ranges more efficiently than the traditional class-based system.
- Each VPC requires a CIDR block to define the range of IP version 4 (IPv4) addresses that resources within the VPC can use.
- Default VPCs have a CIDR of 172.31.0.0/16,
  - IP Portion (172.31.0.0): This is the starting address of the block.
  - Suffix (/16): The “/16” indicates that the first 16 bits of the IP address are fixed, leaving the remaining 16 bits available for host addresses. This provides a range from 172.31.0.0 up to 172.31.255.255
- You must choose a CIDR size between /16 and /28
  - The smaller the CIDR size, the greater the number of IP addresses available to the VPC.
  - Any CIDR you could assign to a traditional network can also be assigned to a VPC.

## Virtual Private Cloud – Subnets

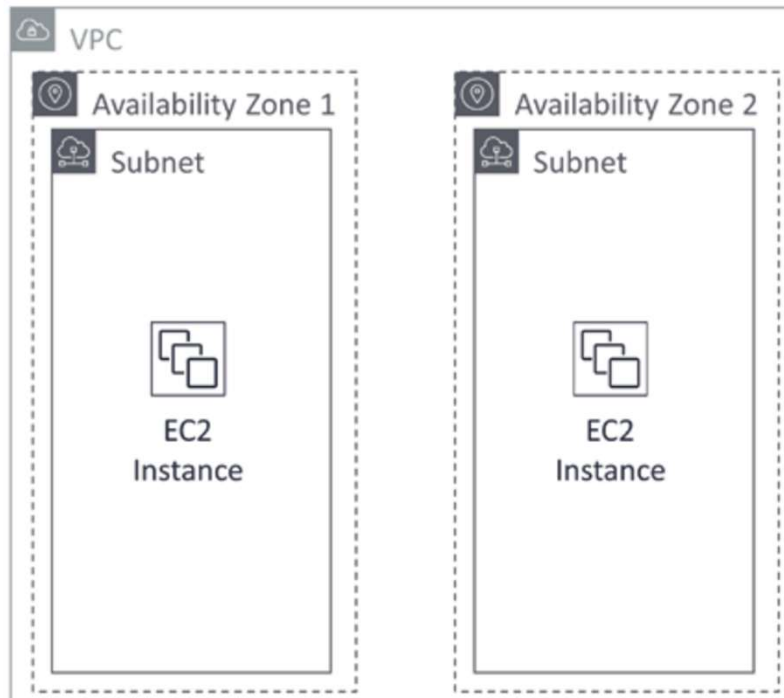
- A VPC is divided into subnets for logical separation and isolation of resources.
  - Subnet CIDR must be a subset of VPC CIDR (/16 to /28).
  - Each subnet exists in a single AZ and contains EC2 instances
- Ranges of IP addresses within a VPC and an AZ. They can be:
  - **Public Subnet** Contains resources that must be connected to the internet. Its route table directs internet-bound traffic to an Internet Gateway.
  - **Private Subnet** Contains sensitive resources like databases. Its route table does not have a direct route to the internet gateway. Instances here can access the internet via a NAT gateway in a public subnet but cannot be accessed from the internet directly.
- **Route Tables** Sets of rules that determine where network traffic from a subnet is directed.

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A VPC with two subnets in different availability zones



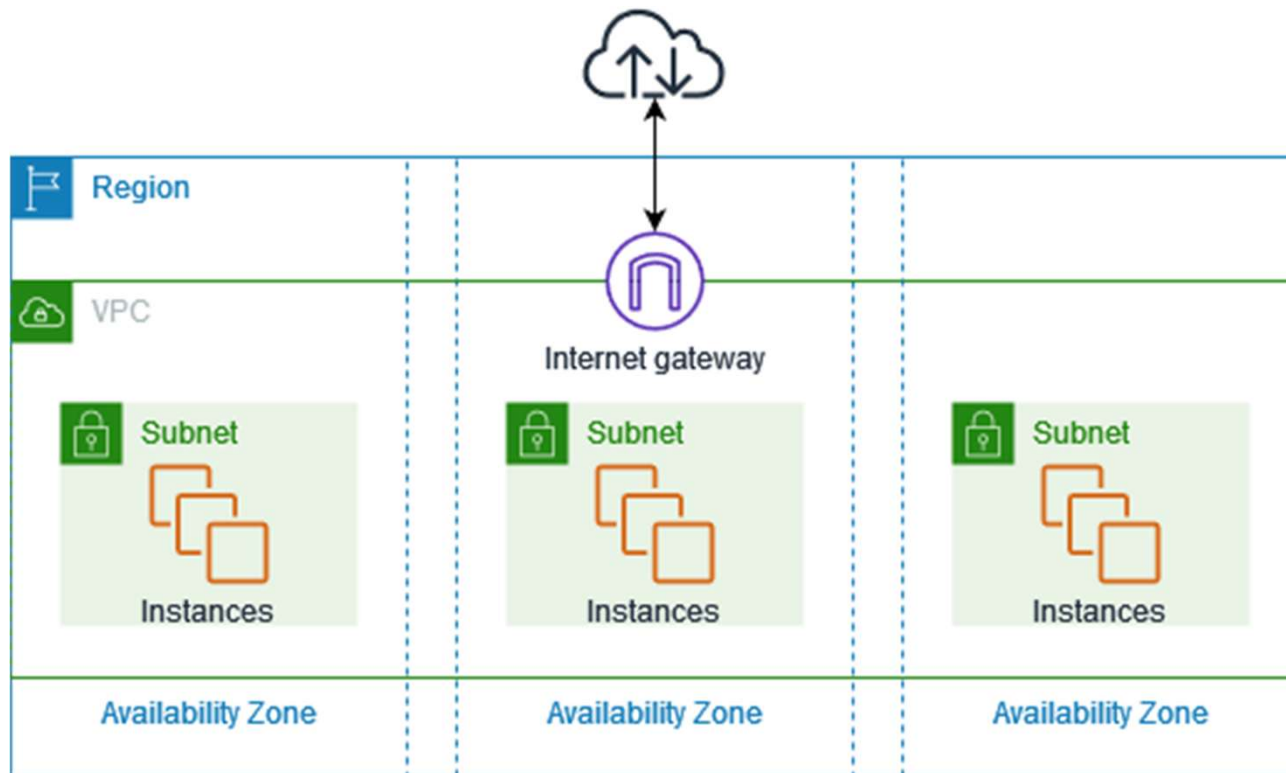
## Virtual Private Cloud – Internet Gateway (IGW)

- An Internet Gateway (IGW) connects a VPC's public subnets to the internet
- An Internet gateway allows EC2 instances in a VPC to access the Internet.
  - Subnets with a default route to an Internet gateway are public subnets.
  - Instances need a public IP (assigned automatically or via Elastic IP) for Internet access
- An Internet Gateway (IGW) is attached **per VPC**, not per subnet
- In summary, the IGW is a VPC-level resource, but its connectivity is configured at the subnet level via route tables.

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## Virtual Private Cloud – Internet Access



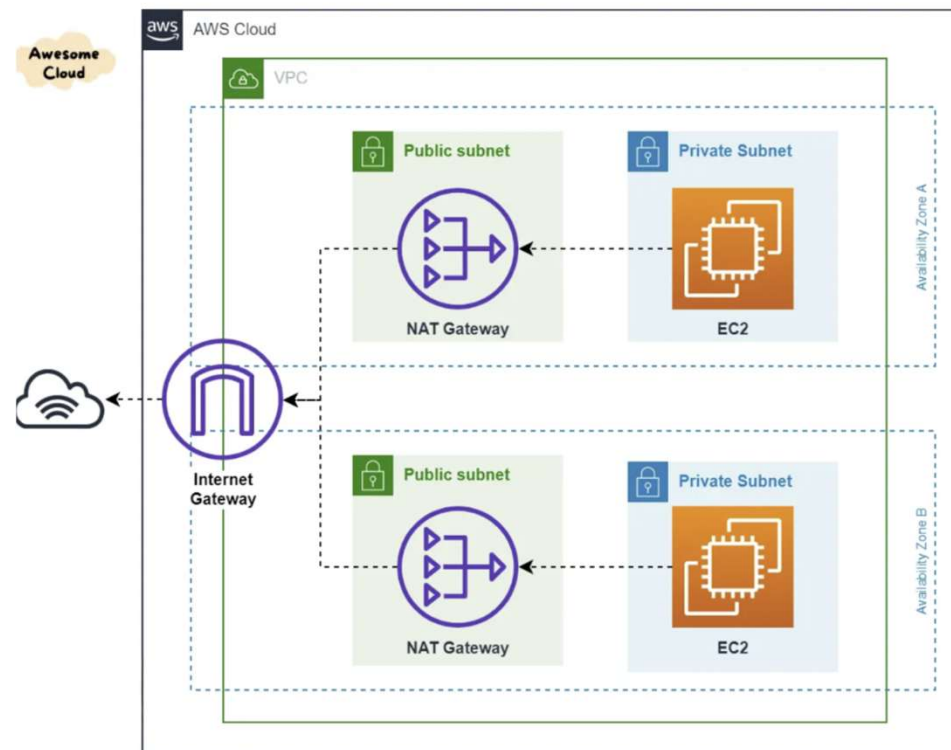
## Virtual Private Cloud – NAT Gateway

- NAT Gateway (NGW) is a managed Network Address Translation (NAT) service.
- NAT Gateway does something similar to Internet Gateway (IGW), but it only works one way: Instances in a private subnet can connect to services outside your VPC but external services cannot initiate a connection with those instances.
- Each NAT gateway is created in a specific Availability Zone and implemented with redundancy in that zone.
- *NAT gateway replaces the source IP address of the instances with the IP address of the NAT gateway.*

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## AWS Networking Services

### Virtual Private Cloud – NAT Gateway



Awesome Cloud — AWS Internet gateway (igw) and NAT gateway (ngw)

## Virtual Private Cloud – Security Groups

- A security group acts as a firewall for instances, controlling inbound and outbound traffic based on IP and protocol.
  - Default: no inbound rules, allows all outbound traffic.
  - Inbound rules control incoming traffic; outbound rules control outgoing traffic
  - No "Deny" Rules: Only "allow" rules can be created. If a rule doesn't explicitly allow traffic, it is denied
- **Stateful Inspection:** If an inbound request is permitted, the corresponding outbound response is automatically allowed, regardless of outbound rules.
- **Instance-Level Security:** Security groups are associated with ENIs (Elastic Network Interfaces) of instances rather than subnets.
- **Multiple Association:** You can assign up to five security groups to a single network interface.



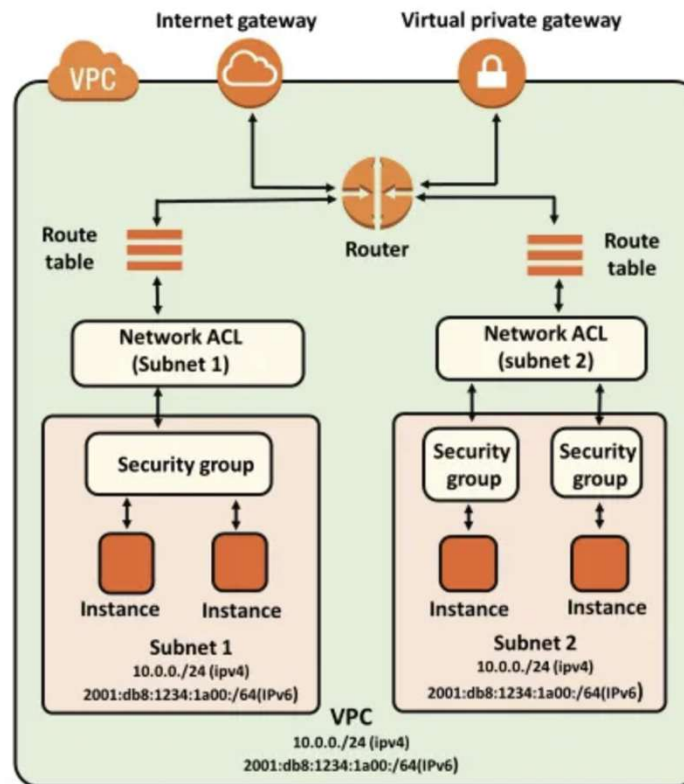
## Virtual Private Cloud – Network Access Control Lists (NACLs)

- AWS Network Access Control Lists (NACLs) are stateless, subnet-level firewalls
  - Default Behavior: Default NACLs allow all traffic, but custom NACLs deny all traffic until configured.
  - Rule Order: Rules are evaluated starting with the lowest numbered rule.
  - Allow/Deny: NACLs support both explicit Allow and Deny rules.
- Stateless: You must explicitly define rules for both incoming traffic and the corresponding outgoing response.
- Subnet Level: Rules apply to all instances within the subnet(s) associated with the NACL
- Association: A NACL can be associated with multiple subnets, a subnet cannot have multiple NACLs.

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### Virtual Private Cloud – SGs vs NACLs



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## Typical Route Table Entries

| Destination    | Target      | Description                                                  |
|----------------|-------------|--------------------------------------------------------------|
| 10.0.0.0/16    | local       | Allows internal communication within the VPC.                |
| 0.0.0.0/0      | igw-xxxxxxx | Sends all internet-bound traffic to the Internet Gateway.    |
| 192.168.1.0/24 | nat-xxxxxxx | Routes traffic for a specific network through a NAT gateway. |



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## Unit 2 - Experiential Learning

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- Launching a free-tier EC2 instance,
- Private and Public VPC Services.

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## EXERCISE 1

### Select an EC2 AMI

1. On the EC2 Dashboard, click the Launch Instance button.
2. Click Browse More AMIs in the Quick Start menu to enable all four tabs. Scroll through the available Quick Start options, noting the various operating systems (Amazon Linux, SUSE Linux, Windows, etc.).
3. Try to identify each one's OS and release number (Ubuntu 22.04, for example), root device type (EBS), AMI ID (for instance, ami- 06ca3ca175f37dd66), and the choice of hardware architecture (in most cases, it will be either 64- bit x86 or 64-bit ARM).

..contd



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#### Select an EC2 AMI (..contd)

4. Now switch to the AWS Marketplace tab, scroll down to the links to specific categories, and click the Business Applications link.
5. Now switch to the Community AMIs tab, scroll down to the links to specific categories, and click the Business Applications link. Look at verified by AWS
6. Unless you want to go ahead and launch a new instance right now—and are aware of any costs you might incur— click Cancel and then Exit.

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## EXERCISE 2

1. On the AWS Management console search bar type vpc
2. Select VPC and go to VPC dashboard
3. On the left pane select 'Your VPCs' under Virtual private cloud
4. You'll find the default VPC (and any other VPCs if you created any)
5. Click on the vpc id.
6. See the subnets and route tables, CIDR allocated to the VPC
7. Go to VPC dashboard and click 'Create VPC' to see the flow and finally cancel.

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## EXERCISE 2

1. On the left pane select 'Subnets' under Virtual private cloud go to subnet dashboard
2. You'll find the default subnets (and others if you created any)
3. Click on the subnet id.
4. See the route tables, CIDR allocated to the subnet
5. Go to Subnet dashboard and click 'Create Subnet' to see the flow and finally cancel.

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## EXERCISE 2

1. Explore Route Tables, Internet gateways and NAT gateways under Virtual private cloud.
2. Explore Network ACLs and Security groups under Security



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